

Fong Shu Hung (Harrison)

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EDUCATION

University of Hong Kong

Hong Kong

Bachelor of Data Science and Engineering, Double Major in Mathematics

Sept 2023 — Expected June 2027

- Mathematics & Statistics: Multivariate data analysis (A), Linear Statistical Analysis (A), Probability & Statistics (A), Optimization, Numerical Analysis, Probability Theory
- Data Science: Natural Language Processing (A+), Machine Learning (A), Statistical Machine Learning (A-), Exploratory Data Analysis (A), Deep Learning, Advanced Algorithm Design

RESEARCH EXPERIENCE

Research Assistant (NLP & Information Retrieval)

June 2026 — Sep 2026

Department of Mathematics, The University of Hong Kong

- Investigated how semantically equivalent query representations affect pretrained code retrieval, using Manim as a controlled domain for studying natural-language-to-program alignment.
- Built a reproducible Scene-level retrieval benchmark from 7,633 Manim Scene classes across eight open-source collections, with structural grouping, dense chunking, frozen queries, and provenance-preserving evaluation artifacts.
- Evaluated BM25 and four pretrained code retrievers across 20 frozen intents and 80 query representations using nDCG@10, paired bootstrap confidence intervals, exact sign-flip tests, and Holm correction; identified model-conditional representation effects rather than consistent gains from structured queries.
- Designed an LLM-assisted graded-relevance protocol calibrated against 354 blind-first human judgments, achieving 84.2% binary agreement and a quadratic-weighted kappa of 0.704.

Research Assistant (Human-Centred AI)

July 2025 — August 2025

Institute of Software, Chinese Academy of Sciences

- Designed and developed a human-centered conversational AI agent to assist human decision-making and provide adaptive interaction through multi-modal and multi-LLM architectures.
- Designed a scalable multi-user interaction architecture enabling AI agents to support collaborative decision-making scenarios with concurrent human inputs.
- Investigated conversational context management challenges in human-AI interaction and developed prompt-based strategies to maintain user intent tracking and dialogue coherence.
- Integrated real-time speech interaction capabilities using Whisper (STT) and CosyVoice (TTS), enabling multimodal communication between users and AI agents.

Research Assistant (RAG & Document Intelligence)

Sep 2025 — Present

IEXIA Limited

- Independently designed and developed an LLM-assisted financial document intelligence pipeline for extracting structured information from reports, supporting rolling reports from nearly 3,000 companies.
- Investigated the challenge of locating relevant pages in long-context documents and developed three progressively improved search strategies, including weighted keyword retrieval, table-of-content guided navigation, and low-resolution page scanning inspired by human reading behavior. Improved page retrieval reliability from 10% to 70% and ultimately 99%.
- Improved pipeline stability by conducting systematic failure analysis, identifying limitations of previous approaches, and refining search strategies to handle diverse annual report structures.
- Analyzed failure distributions across companies and annual report structures, clustering unsuccessful cases to uncover systematic failure modes and guide targeted improvements to the extraction pipeline.

RESEARCH MANUSCRIPTS

Harrison Fong

2026

Representation Effects Are Model-Conditional: Probing Pretrained Code Retrievers with Manim Queries

Spotlight, LP4FM 2026 (NeurIPS Workshop).

INTERNSHIP

Algorithm and Model Development Intern

May 2025 — July 2025

Midjourney China Lab (Youchuan)

- Built a repeatable Python/Playwright anti-piracy workflow to investigate unauthorized services reselling Midjourney access through rebranded interfaces, supporting enforcement against associated accounts and websites.
- Automated authenticated browser flows and collected session-state and network evidence across 13 of 14 targeted services (93%), including sites using QR-code authentication and dynamic frontend workflows.
- Designed a modular, class-based architecture for site-specific extraction logic, enabling additional services to be integrated without modifying the core automation workflow.
- Reverse-engineered JavaScript execution and request-polling behavior to stabilize automation against window-detection, debugger-trigger, and other anti-automation mechanisms.

Agent Researcher & Engineer

Aug 2026 — Present

Hangzhou Hei Ba Meng Qi Ji Media Technology Co., Ltd.

- Translated ambiguous client requirements into a PRD, two-phase delivery roadmap, and modular implementation plan for an agent-enabled WeChat Mini Program covering product copywriting, promotional imagery, digital-human sales videos, and a product-selection marketplace.
- Mentor two interns through weekly planning meetings and demo reviews, decomposing milestones into module-level tasks and assigning ownership based on domain expertise, autonomy, and execution pace while maintaining cross-module knowledge sharing.
- Designed a user-in-the-loop agentic workflow that progresses from product copy generation to optional image generation and digital-human video production, with approval checkpoints between stages to preserve user control.
- Screened 10+ open-source generative-media projects for functional similarity and reuse potential, identifying reference components for video scripting, storyboarding, product-image generation, and e-commerce copywriting.
- Prototyped and tested OpenRouter support within a selected project's provider adapter; diagnosed architectural coupling to a proprietary AI relay platform and scoped the refactoring required for provider-independent model integration.

PROJECTS

Collaborator, Multimodal 30-Day Hospital Readmission Prediction (Kaggle)

Nov 2025 — Dec 2025

- Achieved a validation AUC of 0.8694 by combining EHR sequences, X-ray embeddings, and clinical notes through an AutoGluon weighted ensemble.
- Engineered multimodal features using GRUs for temporal EHR data, PCA for high-dimensional image embeddings, and clinical NLP pipelines incorporating the Charlson Comorbidity Index.
- Eliminated look-ahead bias from the preprocessing pipeline to preserve temporal causality and produce deployment-realistic validation results.

On-Stage Talker, Web Traffic Shifts: Search Engines vs. LLMs

June 2024

- Presented an R/ggplot2 exploratory analysis of changing user behaviour to a technical audience of approximately 100.

EXPERIENCE

Teaching Assistant - COMP2501 Introduction to Data Science

Jan 2025 — Jun 2025

Department of Computer Science, The University of Hong Kong

- Independently designed 50 AI-resistant candidate questions for a 200-student midterm examination, with the explicit goal of preventing students from succeeding through direct or uncritical reliance on LLM-generated answers.
- Constructed image- and data-intensive R programming problems requiring code execution, exploratory analysis, visualization, statistical reasoning, and regular-expression processing; submissions from the TA team were consolidated into a 75-question final assessment.

TECHNICAL SKILLS

- **Programming:** Python (PyTorch, scikit-learn, pandas, NumPy), TypeScript (Next.js, React, Node.js), R (tidyverse, ggplot2), SQL, C/C++, MATLAB
- **AI/ML:** LLM Agents, RAG, Information Retrieval, AutoGluon, OpenRouter, OpenAI-compatible APIs
- **Tools:** Git, Playwright, Vitest, Typst, LaTeX

LANGUAGES:

- Cantonese: Native
- Mandarin: Fluent (written and spoken)
- English: Fluent (written and spoken)

PERSONAL INTEREST

- Interests: Photography (Nikon), Strategic Debate, Creative Writing, Weight Training